William Makinen

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EDUCATION

Princeton University: Master's in Engineering; GPA: 3.7
M.Eng in Mechanical and Aerospace Engineering *May 2023*

Princeton University: Batchelor's of Science in Engineering; GPA: 3.66
B.S.E. in Electrical and Computer Engineering, Computer Science and Robotics Certificates *May 2022*

EXPERIENCE

Mytra Inc, Brisbane, CA *June 2024 – Present*

- Designing, bringing up, and iterating upon new PCBs and wiring harnesses for Mytra's warehouse automation robots.
- Developing internal tools for PCB part library management and custom PLM integrations and workflows.

Brilliant Smart Home, San Mateo, CA *June 2022 - May 2024*
Hardware Team Electrical Engineer

- Improved upon Brilliant's hardware products as core member of the hardware team.
    - Designed, prototyped, and tested PoE variant of Brilliant Control.
    - Debugged critical noise issues within Brillian Switch product.
- Contributed via circuit design, PCB layout, prototyping, and bring-up.

Brilliant Smart Home, San Mateo, CA *Summer 2022*
Hardware Team Intern

- Improved upon Brilliant's hardware products as core member of the hardware team.
    - Researched and implemented doppler radar motion sensor candidates as an alternative to PIR technologies for use in the Brilliant Control.
    - Developed a battery-powered Brilliant Control for use as a demo unit.
- Contributed via circuit design, component selection, prototyping, and PCB layout.

E3D-Online, Oxford, UK *Summer 2021*
Intern

- Designed, developed, and tested FDM 3D printing extrusion systems as part of the engineering team.
- Built internal testing rigs and developed accompanying test procedures for company products and prototypes.

OPEX Corporation, Moorestown, NJ *Summer 2019 & Summer 2020*
Electrical Engineering Intern

- 2020: Worked in Incoming and Scanning Division on variety of circuit design and embedded systems projects.
- *2019*: Worked in Warehouse Automation Division to streamline and automate final QC process for SSXL iBots.

U.S. Naval Research Laboratory, Washington, DC *Summer 2017 & Summer 2018*
Student Researcher

- *2018:* Experimented with the 3D printing of technical ceramics as an alternative fabrication method to previous year's work.
- *2017:* Worked in Optical Sciences division to construct a binder-jetter 3D printer capable of forming technical ceramics into complex shapes.

Intel International Science and Engineering Fair (ISEF), Pittsburgh, PA and Los Angeles, CA
Award Winner *May 2017*

- Developed fully autonomous system for detecting and correcting errors during the 3D printing process. Received 4th place in Engineering and Mechanics category at international fair.
- **US Patent Application Granted:** US Patent No. 11,084,091.

Participant *March 2016*

- Built autonomous, computer vision-controlled object collection robot to seek out and collect designated objects in a room. Intended to help the disabled keep homes clean and safe. Placed 2nd at regional fair.

Finalist *May 2015*

- Constructed a robotic hand designed to assist with recovery process of temporary forearm injuries (broken radii, etc.). Grand prize at Fairfax County regional fair, finalist at international fair.

Virginia Governor's Mentorship in Engineering, NASA Langley Research Center, Hampton, VA *Summer 2016*
Student Researcher
- Worked with other research engineers on real-time 3D printer characterization research to detect print errors in real time. Work expanded upon for 2017 Intel ISEF project.

ACTIVITIES

West Valley Track Club *12-15 hours per week*
Social Chair
- Organize team social activities, help new members feel at home in the club, and assist with recruitment.
- Compete in track, road, and cross country races, attend workouts/practices, and volunteer at club events/races.

SKILLS

Python, C/C++, Java, MATLAB, Linux Systems, Circuit Design and Simulation, PCB Design and Layout, Verilog and FPGA, CAD (SolidWorks, Inventor/Fusion360), G-Code and CNC Devices, Oscilloscopes, DMMs, Signal Generators, Workshop Tools and General Making.